At-Compromise Security: The Case for Alert Blindness

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Martin R. Albrecht[1], Simone Colombo[1], Benjamin Dowling[1] and Rikke Bjerg Jensen[2]

[1]King's College London, [2]Royal Holloway, University of London

Content Warning

We talk about abductions and torture.

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- "[...] **we believe faithfully models** the adversarial behavior in many commercial, political, and social settings."

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- "A particular **example of this model in the real world** is a dictator providing citizens with electronic identity documents […] and blocking all other usage of cryptography."
- "[…] **we believe faithfully models** the adversarial behavior in many commercial, political, and social settings."

> *But where do these requirements come from? How does one identify such "natural security concerns"?* ***On these questions, the literature remains silent.*** [Bla12]

Cryptography is also a **social science** unaware of itself.

Our answer: ethnography

1. Through **ethnographic findings** we analyse **social relations**,
2. to **discover security concerns** and study them with **cryptography**.

Outline

1 Ethnography

2 Social foundations

3 At-compromise alert blindness

4 Conclusion and future work

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Context: Gen-Z protests in Kenya

- Unprecedented **large-scale protests** in 2024, sparked by the 2024 **Finance Bill**.
- **25 June 2024**: protesters breached the fence around Kenya's Parliament, some entered Parliament and the Senate. State forces killed and injured protesters.
- The 2024 Finance Bill was **withdrawn on 26 June**; smaller-scale protests until 8 August.

Context: abductions

Abductions by William Ruto's repressive forces emerged as a new threat during **these** protests.

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- June-December 2024: **82 abductions** [Ken24].
- 25 June 2024: **coordinated abductions** of at least **50** (known) protesters.
- Additional **15** abductions linked to the one-year anniversary protests [Ken25].

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The **whereabouts** of several abducted people remain **unknown**.

Ethnographic methodology

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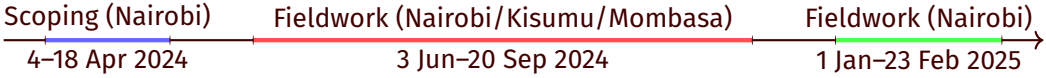
It is through the application of this technique that we arrived at our new security notion.

Fieldwork

Long immersion in a social movement in Kenya that opened access to a wide network of activists.



Data collection: deep hanging out [Gee98]

Active participant observation

Ethnographic interviews in the field

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Active participant observation

- 11 protests in Nairobi.
- Activist meetings, planning sessions, strategic discussions, community gatherings.
- Day-to-day activities in informal settlements (community outreach, hospital and prison visits, group meetings)

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Ethnographic interviews in the field

- 77 protesters (organisers, frontliners, first-timers, seasoned; 8 abducted).
- Loosely structured around a topic guide.
- Duration: 43 minutes to over two hours.

Data analysis

- Analysis across ethnography and cryptography: during, between, after fieldwork.

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Data analysis

- Analysis across ethnography and cryptography: during, between, after fieldwork.
- **Inductive reasoning**: theory developed from empirical data, not from hypotheses.
- We did not test an intuited security notion, but analysed the data to **discover** it.
- Initial analysis by fieldworker, followed by collective analysis and formalisation with cryptographers and fieldworker.

Autumn school on the Social Foundations of Cryptography

https://social-foundations-of-cryptography.gitlab.io/school

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The contours of abductions: interrogation

***[I]t's psychological torture**, because they [interrogators] tell you: 'Just remember, you are not going back [...] so it depends on how you **answer us** [...]. Who is financing you? What are you planning?' [...] They were using a tool. **I don't know whether it was an iron or wood, hitting me on the arm.** So, I still have injuries [rolling up their sleeve to show the scars].* (P72, abductee, Nairobi)

Raising the alarm

During the Gen-Z protests, the practice of ‘raising the alarm’ emerged in response to abductions.

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> *[…] I wrote to him a message. Very brief message: ‘Hey, I’m being abducted. I don’t know where I am’ […] After I had texted him, he took the screenshot [of the message] and posted it on X. So that is how it spread […] funnily enough, after like 30 minutes or so, as they [the abductors] were driving me around, the car stopped again. […] They opened the trunk, and then they started beating me up. And then they asked me: ‘Where is that phone?’ So they knew that I had another phone because my friend shared my screenshot […]* ***I think that is the only thing that saved me. Because if it wasn’t for that message, these guys had another motive.*** *Because nobody was aware of my abduction, apart from that one message. So they could have killed me and taken me somewhere, nobody could have known.* (P51, abductee, Nairobi)

Shouting!

Posters calling for the release of abducted protesters.

- Once the alarm had been raised, shouting could happen across platforms.

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Posters calling for the release of abducted protesters.

- Once the alarm had been raised, shouting could happen across platforms.
- A network of connected and known activists amplified the message.
- This created a feedback loop: the abductors eventually heard the message, which put pressure on them to release the abducted person.

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At compromise alert blindness

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	$\xleftarrow{\mathsf{sm}}$	
$(\cdot, \cdot, k) \leftarrow \mathsf{GetKey}(\mathsf{ds}, \mathsf{sm})$		

Security notion: KIND-AtC-AB

Game KIND-AtC-AB

$\mathsf{ps}_1 \xleftarrow{\mathrm{R}} \mathcal{S}$; $\mathsf{ps}_0 \xleftarrow{\mathrm{R}} \mathcal{S} \setminus \{\mathsf{ps}_1\}$; $k_1, k_0 \xleftarrow{\mathrm{R}} \mathcal{K}$

$(\mathsf{ds}, \mathsf{ss}) \leftarrow \mathsf{GeneralSetup}(1^\lambda, k_1, \mathsf{ps}_1)$

$\mathsf{win}_{\mathsf{cm}}, \mathsf{win}_{\mathsf{sm}} \leftarrow \mathsf{false}$; $d \xleftarrow{\mathrm{R}} \{0,1\}$; $b \xleftarrow{\mathrm{R}} \{0,1\}$

$\sigma \leftarrow \mathcal{A}^{\textsc{Request},\textsc{Response},\textsc{GetKey},\textsc{Clear}}$

if $\mathsf{win}_{\mathsf{cm}} \vee \mathsf{win}_{\mathsf{sm}}$: **return** 1

$\mathsf{ds} \leftarrow \mathsf{Clear}(\mathsf{ds})$

$\mathsf{detect}, \mathsf{received} \leftarrow \mathsf{false}$

$(b', d') \leftarrow \mathcal{A}^{\textsc{Detect},\textsc{Enc}}(\sigma, \mathsf{ds}, \mathsf{ps}_b)$

if $d = d' \wedge \neg\mathsf{received}$: **return** 1

if $b = 0 \wedge \mathsf{received} \wedge \neg\mathsf{detect}$: **return** 1

if $b = b'$: **return** 1

return 0

Oracle $\textsc{Request}()$

$(\mathsf{ds}, \mathsf{cm}) \leftarrow \mathsf{Request}(\mathsf{ds}, \mathsf{ps}_1)$

$\mathcal{C} \leftarrow \mathcal{C} \cup \{\mathsf{cm}\}$

return cm

Oracle $\textsc{Response}(\mathsf{cm})$

$(\mathsf{acc}, \mathsf{ss}, \cdot, \mathsf{sm}) \leftarrow \mathsf{Response}(\mathsf{ss}, \mathsf{cm})$

if $\mathsf{acc} \wedge \mathsf{cm} \notin \mathcal{C}$: $\mathsf{win}_{\mathsf{cm}} \leftarrow \mathsf{true}$

$\mathcal{Z} \leftarrow \mathcal{Z} \cup \{\mathsf{sm}\}$

return sm

Oracle $\textsc{GetKey}(\mathsf{sm})$

$(\mathsf{acc}, \mathsf{ds}, \cdot) \leftarrow \mathsf{GetKey}(\mathsf{ds}, \mathsf{sm})$

if $\mathsf{acc} \wedge \mathsf{sm} \notin \mathcal{Z}$: $\mathsf{win}_{\mathsf{sm}} \leftarrow \mathsf{true}$

Oracle $\textsc{Detect}(\mathsf{cm})$

$(\mathsf{acc}, \mathsf{ss}, \mathsf{detect}', \mathsf{sm}) \leftarrow \mathsf{Response}(\mathsf{ss}, \mathsf{cm})$

if $\neg\mathsf{acc}$: **return** $\perp$

$\mathsf{received} \leftarrow \mathsf{true}$

$\mathsf{detect} \leftarrow \mathsf{detect} \vee \mathsf{detect}'$

return sm

Oracle $\textsc{Clear}()$

$\mathsf{ds} \leftarrow \mathsf{Clear}(\mathsf{ds})$

Security notion: KIND-AtC-AB

Claim 1 *Consider* KIND-AtC-AB *as defined in Definition 4. Model "confidentiality" as* KIND *security. Then* KIND-AtC-AB *correctly responds to security needs of some organisers and frontliners in Kenya in 2024.*

Game KIND-AtC-AB

$\mathsf{ps}_1 \xleftarrow{R} \mathcal{S}$; $\mathsf{ps}_0 \xleftarrow{R} \mathcal{S} \setminus \{\mathsf{ps}_1\}$; $k_1, k_0 \xleftarrow{R} \mathcal{K}$

$(\mathsf{ds}, \mathsf{ss}) \leftarrow \mathsf{GeneralSetup}(1^\lambda, k_1, \mathsf{ps}_1)$

$\mathsf{win}_{\mathsf{cm}}, \mathsf{win}_{\mathsf{sm}} \leftarrow \mathsf{false}$; $d \xleftarrow{R} \{0,1\}$; $b \xleftarrow{R} \{0,1\}$

$(b', d') \leftarrow \mathcal{A}^{\mathsf{Request}, \mathsf{Detect}, \mathsf{GetKey}}(0, \mathsf{ds}, \mathsf{ps}_b)$

if $d = d' \wedge \neg\mathsf{received}$: **return** 1

if $b = 0 \wedge \mathsf{received} \wedge \neg\mathsf{detect}$: **return** 1

if $b = b'$: **return** 1

return 0

Oracle Request()

$(\mathsf{ds}, \mathsf{cm}) \leftarrow \mathsf{Request}(\mathsf{ds}, \mathsf{ps}_1)$

$\mathcal{C} \leftarrow \mathcal{C} \cup \{\mathsf{cm}\}$

return cm

Oracle Detect(cm)

$(\mathsf{acc}, \mathsf{ss}, \mathsf{detect}', \mathsf{sm}) \leftarrow \mathsf{Response}(\mathsf{ss}, \mathsf{cm})$

if $\neg\mathsf{acc}$: **return** $\perp$

$\mathsf{received} \leftarrow \mathsf{true}$

return sm

Oracle GetKey(sm)

$(\mathsf{acc}, \mathsf{ds}, \cdot) \leftarrow \mathsf{GetKey}(\mathsf{ds}, \mathsf{sm})$

if $\mathsf{acc} \wedge \mathsf{sm} \notin \mathcal{Z}$: $\mathsf{win}_{\mathsf{sm}} \leftarrow \mathsf{true}$

Realism: personal secrets

Device state ds and personal secret ps are **different** things.

Game KIND-AtC-AB

$\mathsf{ps}_1 \xleftarrow{\mathrm{R}} \mathcal{S}$; $\mathsf{ps}_0 \xleftarrow{\mathrm{R}} \mathcal{S} \setminus \{\mathsf{ps}_1\}$; $k_1, k_0 \xleftarrow{\mathrm{R}} \mathcal{K}$

[...]

$(b', d') \leftarrow \mathcal{A}^{\textsc{Detect},\textsc{Enc}}(\sigma, \mathsf{ds}, \mathsf{ps}_b)$

if $d = d' \wedge \neg\mathsf{received}$: **return** 1

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- Personal secrets capture the **distinction of compromising people or their devices**.

Realism: personal secrets

Person can **decide** which personal secret to use.

Game KIND-AtC-AB

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$(b', d') \leftarrow \mathcal{A}^{\textsc{Detect},\textsc{Enc}}(\sigma, \mathsf{ds}, \mathsf{ps}_b)$

if $d = d' \wedge \neg\mathsf{received}$: **return** 1

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- Personal secrets capture the **distinction of compromising people or their devices**.
- Multiple personal secrets model that the abducted person knows they have been abducted: **they can decide which secret to use**.

Realism: confidentiality

Game KIND-AtC-AB	Oracle DETECT(cm)
[...]	$(\mathsf{acc}, \mathsf{ss}, \mathsf{detect}', \mathsf{sm}) \leftarrow \mathsf{Response}(\mathsf{ss}, \mathsf{cm})$
$(b', d') \leftarrow \mathcal{A}^{\mathsf{DETECT},\mathsf{ENC}}(\sigma, \mathsf{ds}, \mathsf{ps}_b)$	**if** $\neg\mathsf{acc}$: **return** $\perp$ AEAD rejection
if $d = d' \wedge \neg\mathsf{received}$: **return** 1	$\mathsf{received} \leftarrow \mathsf{true}$
[...]	$\mathsf{detect} \leftarrow \mathsf{detect} \vee \mathsf{detect}'$
return 0	**return** sm

Realism: confidentiality

Game KIND-AtC-AB	Oracle $\textsc{Detect}(\mathsf{cm})$
$[\ldots]$	$(\mathsf{acc}, \mathsf{ss}, \mathsf{detect}', \mathsf{sm}) \leftarrow \mathsf{Response}(\mathsf{ss}, \mathsf{cm})$
$(b', d') \leftarrow \mathcal{A}^{\textsc{Detect},\textsc{Enc}}(\sigma, \mathsf{ds}, \mathsf{ps}_b)$	**if** $\neg\mathsf{acc}$: **return** $\perp$
if $d = d' \wedge \neg\mathsf{received}$: **return** 1	$\mathsf{received} \leftarrow \mathsf{true}$
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return 0	**return** sm

- *"They wanted to know who was funding these protest."* (P49)

Realism: confidentiality

Adversary wins if they recover KIND bit d **without** sending a valid client message.

Game KIND-AtC-AB	Oracle DETECT(cm)
[...]	$(\mathrm{acc}, \mathrm{ss}, \mathrm{detect}', \mathrm{sm}) \leftarrow \mathrm{Response}(\mathrm{ss}, \mathrm{cm})$
$(b', d') \leftarrow \mathcal{A}^{\mathrm{DETECT,ENC}}(\sigma, \mathrm{ds}, \mathrm{ps}_b)$	**if** $\neg\mathrm{acc}$: **return** $\perp$
if $d = d' \wedge \neg\mathrm{received}$: **return** 1	$\mathrm{received} \leftarrow \mathrm{true}$
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return 0	**return** sm

- *"They wanted to know who was funding these protest."* (P49)
- We exploit the abductors' goal to access information: breaking KIND requires **at least** one client's message to reach the server ($\mathrm{received} \leftarrow \mathrm{true}$).

Realism: alert

Game KIND-AtC-AB	Oracle $\textsc{Detect}(\mathsf{cm})$
$\mathsf{ps}_1 \xleftarrow{R} \mathcal{S}$; $\mathsf{ps}_0 \xleftarrow{R} \mathcal{S} \setminus \{\mathsf{ps}_1\}$; $k \xleftarrow{R} \mathcal{K}$	$(\mathsf{acc}, \mathsf{ss}, \mathsf{detect}', \mathsf{sm}) \leftarrow \mathsf{Response}(\mathsf{ss}, \mathsf{cm})$
$[\ldots]$	**if** $\neg\mathsf{acc}$: **return** $\perp$
$(b', d') \leftarrow \mathcal{A}^{\textsc{Detect},\textsc{Enc}}(\sigma, \mathsf{ds}, \mathsf{ps}_b)$	$\mathsf{received} \leftarrow \mathsf{true}$
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Realism: alert

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Make **disappearance publicly known**: *"I think that is the only thing that saved me."* (P51)

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[...]	

Adversary wins if given **any random** $\mathsf{ps}_0 \xleftarrow{R} \mathcal{S}$ different from the setup personal secret ps_1 they manage to **not** trigger an alarm.

Make **disappearance publicly known**: *"I think that is the only thing that saved me."* (P51)

Realism: blindness

Game KIND-AtC-AB

$b \xleftarrow{\mathrm{R}} \{0, 1\}$

[...]

$(b', d') \leftarrow \mathcal{A}^{\textsc{Detect},\textsc{Enc}}(\sigma, \mathrm{ds}, \mathrm{ps}_b)$

[...]

if $b = b'$: **return** 1

return 0

Client		**Server**
		$(\cdot, \cdot, \mathrm{detect}, \mathrm{sm}) \leftarrow \mathrm{Response}(\mathrm{ss}, \mathrm{cm})$
	$\xleftarrow{\mathrm{sm}}$	
$(\cdot, \cdot, k) \leftarrow \mathrm{GetKey}(\mathrm{ds}, \mathrm{sm})$		

- **Non-cooperation**: *"If I do anything now to upset these guys, **I'm dead**."* (P60)

Realism: blindness

Game KIND-AtC-AB

$b \xleftarrow{R} \{0,1\}$

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The protocol **always** reconstructs the key.

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- The protocol **always** reconstructs the key: abducted persons would cooperate regardless.

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- We **give up confidentiality for alert**: a goal **only ethnography could suggest**.

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This notion is achievable: we give a provably secure construction and we implement it.

Outline

1 Ethnography

2 Social foundations

3 At-compromise alert blindness

4 Conclusion and future work

Conclusion and future work

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We apply this technique to study **at-compromise** security.

Ethnographic fieldwork with protesters in Kenya reveals the security goal of **alert blindness**.

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https://simonecolombo.info | simone.colombo@kcl.ac.uk | https://social-foundations-of-cryptography.gitlab.io

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